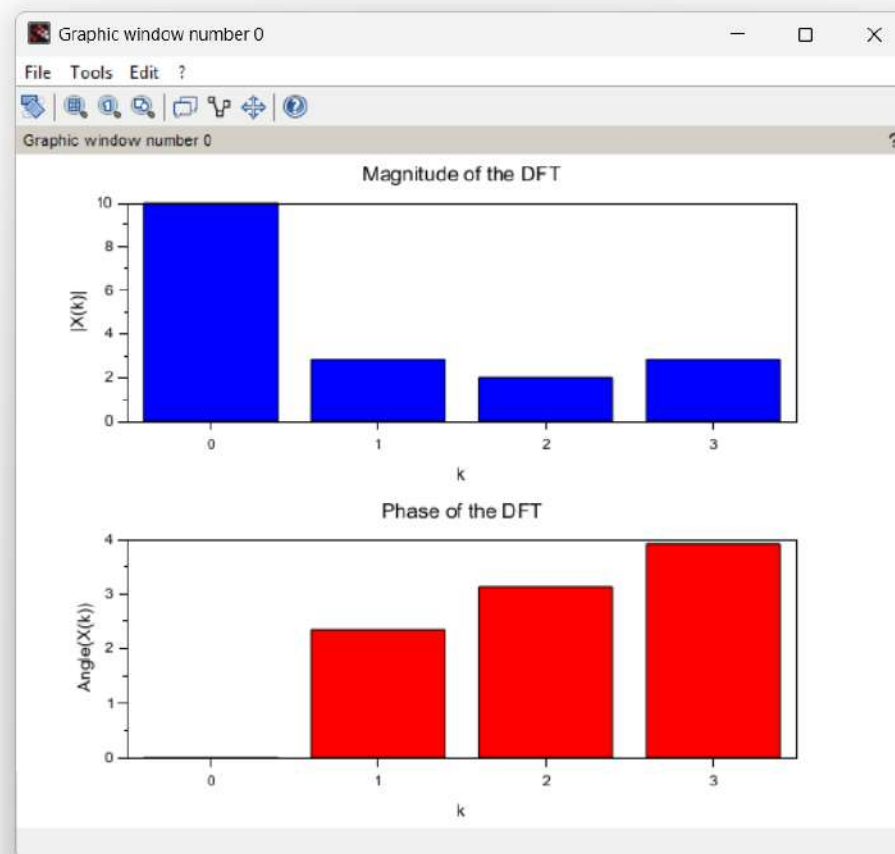


```

12
13 // Initialize arrays to store magnitude and phase
14 magnitude = abs(X_k); // Compute magnitude of DFT
15 phase = zeros(1, N); // Initialize phase array
16
17 // Compute phase manually using atan
18 for i = 1:N
19     real_part = real(X_k(i));
20     imag_part = imag(X_k(i));
21
22     if real_part > 0 then
23         phase(i) = atan(imag_part / real_part); // Phase in [-pi/2, pi/2]
24     elseif real_part < 0 then
25         phase(i) = atan(imag_part / real_part) + %pi; // Adjust phase for left-half-plane
26     else
27         if imag_part > 0 then
28             phase(i) = %pi/2; // Positive imaginary axis
29         elseif imag_part < 0 then
30             phase(i) = -%pi/2; // Negative imaginary axis
31         else
32             phase(i) = 0; // Origin case
33         end
34     end
35 end
36
37 // Plot the magnitude and phase of the DFT
38 clf; // Clear the current figure
39
40 // Plot magnitude
41 subplot(2,1,1);
42 bar(k, magnitude, 'b');
43 xlabel('k');
44 ylabel('|X(k)|');
45 title('Magnitude of the DFT');
46
47 // Plot phase
48 subplot(2,1,2);
49 bar(k, phase, 'r');
50 xlabel('k');
51 ylabel('Angle(X(k))');
52 title('Phase of the DFT');
53

```



exp 9.3.sci (C:\Users\imjan\OneDrive\Documents\exp 9.3.sci) - SciNotes

File Edit Format Options Window Execute ?



exp 9.3.sci (C:\Users\imjan\OneDrive\Documents\exp 9.3.sci) - SciNotes

1.1.exp.sci X 1.2.exp.sci X exp 2.1.sci X 2.2.sci X exp 3.2.sci X
3.1.sci X 4.1.exp.sci X exp 4.2.sci X exp 5.1.sci X exp 5.2.sci X exp 6.1.sci X exp 6.2.sci X exp 7.1.sci X exp 7.3.sci X exp 8.3.sci X exp 8.4.sci X exp 9.1.sci X exp 9.3.sci X

```
1 clc;
2 clear;
3 // Input sequence (length should be power of 2)
4 x = [1, 2, 3, 4, 0, 0, 0, 0];
5 N = length(x);
6 X = fft(x, -1); // Scilab built-in FFT (DIT equivalent)
7 // Display results
8 disp("FFT-(DIT)-Result:");
9 disp(X);
10 subplot(2,1,1);
11 plot(0:N-1, abs(X), 'r');
12 xlabel('Frequency Index');
13 ylabel('Magnitude');
14 title('Magnitude Spectrum (DIT FFT)');
15 subplot(2,1,2);
16 phaseX = atan(imag(X), real(X)); // Phase in radians
17 plot(0:N-1, phaseX, 'b');
18 xlabel('Frequency Index');
19 ylabel('Phase (radians)');
20 title('Phase Spectrum (DIT FFT)');
21
```

